

Analysis and Comparison of Secure State Estimation Algorithms for Cyber-Physical Systems

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Abstract—This paper addresses the problem of secure state estimation for Cyber-Physical Systems (CPS) subject to sparse, constant sensor attacks. We implement and evaluate the performance of four online optimization algorithms: Online Proximal Gradient Descent (O-PGD), Online Proximal Alternating Minimization (O-PAM), Aggregated Online PGD (AO-PGD), and Aggregated Online PAM (AO-PAM). Mathematical formulation establishes that the presence of system eigenvalues at unity causes standard Luenberger observers to fail due to structural unobservability. We demonstrate how sparse optimization schemes successfully decouple state dynamics from persistent corruption vectors, ensuring real-time trajectory tracking and precise attack support identification.

Index Terms—Cyber-Physical Systems, Secure State Estimation, Luenberger Observer, Sparse Recovery, Proximal Optimization.

I. ANALYSIS OF THE SYSTEM DYNAMICS AND ATTACK OBSERVABILITY

Initially, we loaded the provided system data, specifically the $n \times n$ square matrix A , which governs the system dynamics. As a preliminary step to understand the nominal behavior of the system prior to any estimation attempt, we analyzed the unforced dynamics $x(k+1) = Ax(k)$ by computing the eigenvalues of A .

The analysis revealed that all eigenvalues have a magnitude less than or equal to one. In particular, we identified an eigenvalue at exactly one with an algebraic multiplicity of 2. By verifying that the geometric multiplicity is equal to 2, hence matching the algebraic one, we can conclude that the attack-free system is marginally stable. Establishing this nominal baseline is a fundamental methodological prerequisite. Analyzing the unforced dynamics first ensures that the system is inherently well-behaved, guaranteeing that any subsequent estimation challenge is strictly attributable to the malicious injection rather than a pre-existing structural instability.

Subsequently, we analyzed the system's observability using the pair of matrices (A, C) , where C denotes the provided output matrix. By computing the observability matrix $\mathcal{O}$ and verifying that it has full rank, we can safely conclude that the nominal, attack-free system is completely observable.

Following this, we proceeded to introduce a constant attack vector a into the output equation, modeled as $y(k) = Cx(k) + a$, to analyze its impact on the CPS. Crucially, we assume that a is sparse, meaning no more than h components are non-zero, to reflect a realistic threat model representing a malicious actor with limited resources. Under these conditions, we constructed an augmented observability matrix $\mathcal{O}'_T \in \mathbb{R}^{qT \times (n+q)}$, with $qT > n + q$. We then evaluated whether this augmented

matrix is again full column rank ($\text{rank}(\mathcal{O}'_T) = n + q$). This condition is theoretically equivalent to determining whether a unique solution for the system exists, which is necessary to successfully reconstruct the attack. However, it is a known result that if matrix A possesses an eigenvalue $\lambda = 1$, the constant attack dynamics become unobservable because the matrix $(I - A)$ is not full rank. Consequently, since we found two eigenvalues $\lambda = 1$, the system yields infinite solutions, making it impossible to uniquely determine the attack vector.

This mathematical limitation has a direct physical interpretation. If the nominal system were strictly stable, its unforced dynamics would naturally decay, making it easier for an estimator to track the state. However, the presence of the eigenvalue $\lambda = 1$ means the natural unforced dynamics perfectly overlap with the implicit dynamics of a constant attack. Consequently, a standard observer cannot distinguish whether a variation in the measured output $y(k)$ is due to the state moving along the marginally stable mode or if it is caused by the persistent attack.

To perform online state estimation, an approach that computes estimates immediately, avoiding the need to collect every possible measurement of the dataset, one typical approach consists of exploiting a Luenberger, which can accurately track the state estimation error. However, as previously established, the system under a constant attack is unobservable, rendering this classical observer unfeasible.

Alternatively, we must exploit optimization-based algorithms capable of performing online secure state estimation: O-PGD, O-PAM, AO-PGD, and AO-PAM. These algorithms effectively replace the Luenberger observer by producing robust approximations through the minimization of a cost function. Their results identify the solution among the infinite set of possibilities by identifying the one with the lowest number of strictly non-zero terms, thereby leveraging the sparsity of the attack vector. Consequently, rather than attempting to find a unique analytical solution, these algorithms identify the optimal approximation from an infinite space of feasible solutions.

II. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

A. Controlled Failure Analysis (Run 1)

The initial testing phase evaluated the response of the O-PGD, O-PAM, AO-PGD, and AO-PAM algorithms to a deliberately severe and unbalanced parameterization. The objective was to assess the impact of parameter tuning on estimator robustness and detection capability. Specifically, an

excessively penalizing threshold $\lambda = 2.0$ was assigned to all models, operating in direct opposition to overly large descent step sizes.

Table I summarizes the parameter configuration adopted for this evaluation, where ν represents the step size for the PGD-based schemes and μ denotes the regularization weight for the PAM-based methods.

TABLE I: Parameter Configuration for Controlled Failure (Run 1)

Algorithm	μ/ν	λ
O-PGD	0.8	2.0
O-PAM	5.0	2.0
AO-PGD	1.0	2.0
AO-PAM	2.0	2.0

1) *Error Classification (False Negatives and False Positives)*: Assigning an excessive threshold forces the proximal mapping operator to set the estimated vector components to zero. Consequently, the algorithm performs excessive shrinkage, resulting in the misclassification of active attack channels as background noise.

As illustrated in Fig. 1 and Fig. 2, the False Negative metric remains saturated at a constant value of four across all iterations, rendering the monitoring architecture completely blind to physical attacks. Conversely, False Positives are immediately driven to zero due to the extreme mathematical filtering imposed by the regularization weight.

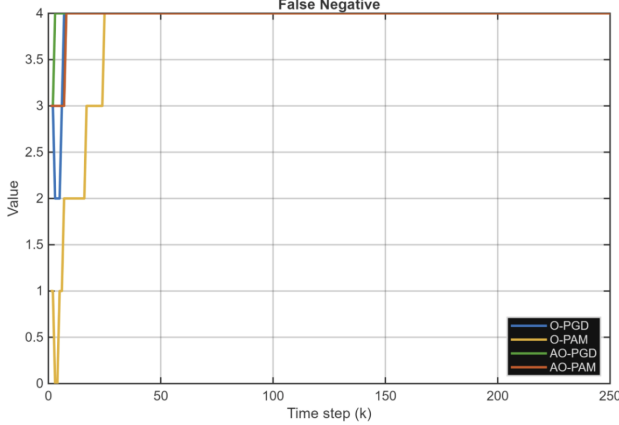


Fig. 1: Run 1 False Negative error across iterations.

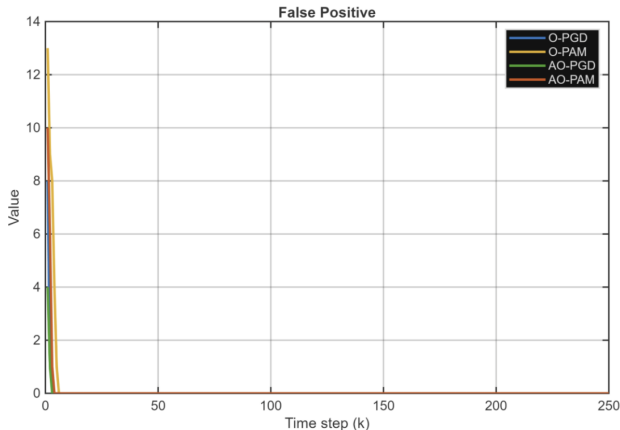


Fig. 2: Run 1 False Positive count across iterations.

2) *Support Attack Error*: Consistent with the misclassification of compromised channels, the Support Attack Error remains saturated at its maximum level. Due to the excessive suppression of the sparse attack term induced by λ , the discrepancy between the true attack support and the estimated support does not undergo any iterative decrease. Even aggressive descent step sizes, such as the parameter $\mu = 5.0$ assigned to O-PAM, prove mathematically ineffective and incapable of overcoming the optimization barrier imposed by the proximal penalty function.

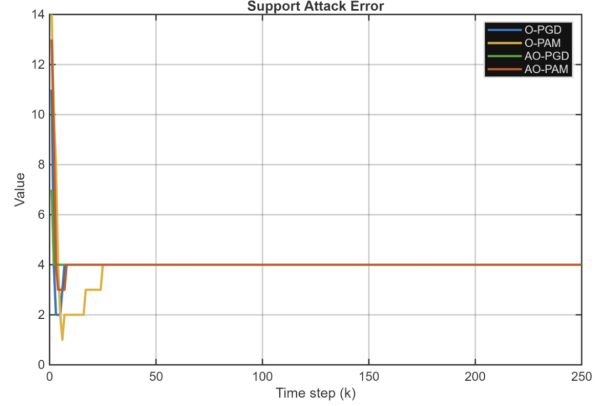


Fig. 3: Run 1 Support Attack Error across iterations.

3) *State Estimation Error*: The inability to identify the attack vector directly induces a critical increase in the State Estimation Error. Deprived of the essential correction provided by the exact recovery of the sparse corruption vector $\mathbf{a}$, the state estimator fails to reconstruct and track the nominal system dynamics. This behavior confirms the operational definition of a controlled failure, demonstrating the impossibility of achieving state estimation with regularization weights of this magnitude.

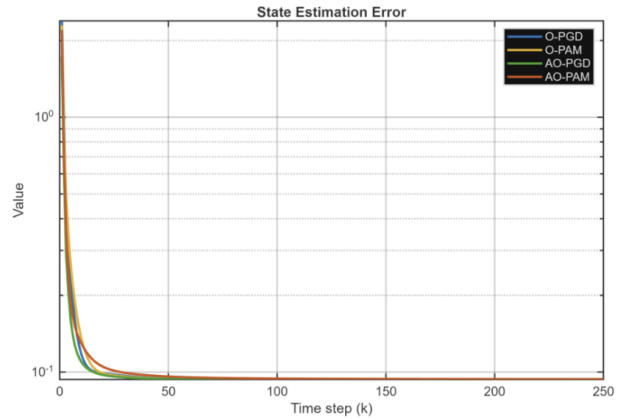


Fig. 4: Run 1 State Estimation Error across iterations.

B. Transition and Underperformance Analysis (Run 2)

In the second experimental run, both the regularization threshold and the descent step size parameters were lowered to recover detection capabilities. However, the configuration remained numerically unbalanced: the assigned step size

proved insufficient to effectively overcome the optimization penalization imposed by the soft-thresholding operator.

Table II outlines the parameterization adopted for this transitional scenario.

TABLE II: Parameter Configuration for Transitional Phase (Run 2)

Algorithm	μ/ν	λ
O-PGD	0.4	1.5
O-PAM	2.0	1.0
AO-PGD	0.5	1.5
AO-PAM	1.0	1.5

1) *Error Classification (False Negatives and False Positives)*: In this transitional scenario, the system successfully suppresses False Positives. However, critical performance limitations persist regarding the complementary metric: False Negatives fail to achieve rapid asymptotic convergence to zero.

This numerical inefficiency is clearly observed in the False Negative profile of the AO-PAM algorithm (Fig. 5), which demonstrates the estimator inability to fully resolve the attack support, stabilizing at a permanent residual error value of one. Conversely, the False Positive count (Fig. 6) confirms complete noise rejection across all algorithmic variants.

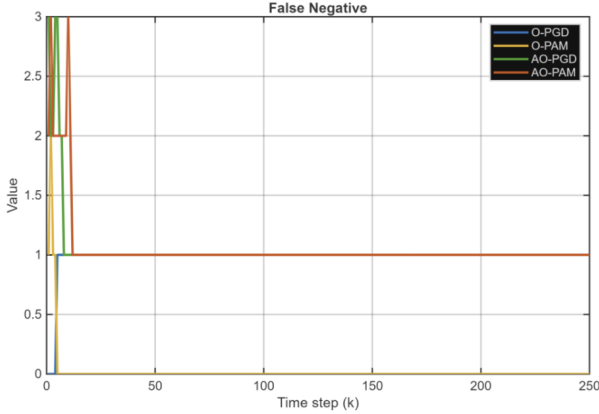


Fig. 5: Run 2 False Negative error across iterations.

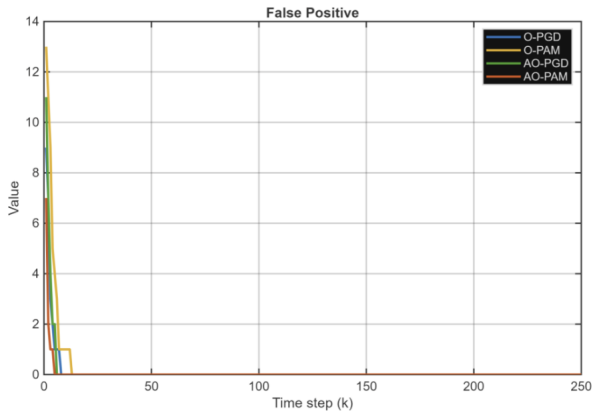


Fig. 6: Run 2 False Positive count across iterations.

2) *Support Attack Error*: The Support Attack Error profile represents the primary indicator of parameter miscalibration. As depicted in Fig. 7, convergence proceeds at an extremely

slow rate, characterized by an evident, repetitive stepped trajectory. This phenomenon indicates that the soft-thresholding operator requires an excessive number of iterations to progressively zero the uncorrupted state components, causing the optimization path to temporarily stall at horizontal local minima before executing subsequent convergence steps.

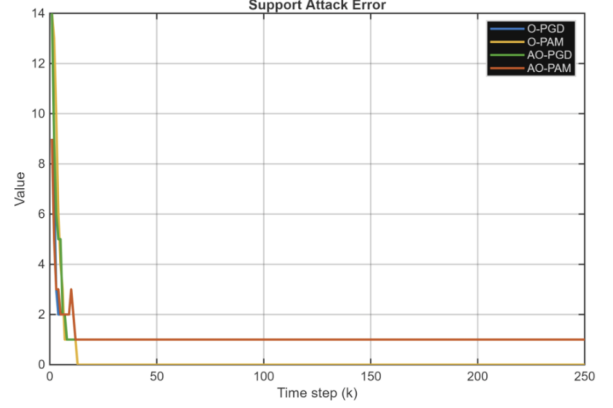


Fig. 7: Run 2 Support Attack Error across iterations.

3) *State Estimation Error*: Due to the extended iteration count required for support identification and the persistence of the stepped convergence profile, the minimization of the State Estimation Error (Fig. 8) is inevitably impeded. The inability to guarantee continuous online tracking degrades the fundamental Secure State Estimation objective of the Cyber-Physical System. Consequently, this run exhibits clear suboptimal tracking performance, providing the analytical rationale for the rigorous parameter balancing implemented in Run 3.

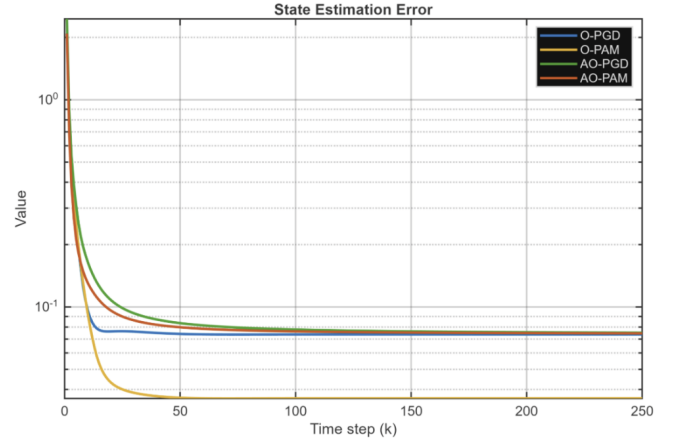


Fig. 8: Run 2 State Estimation Error across iterations.

C. Baseline Balancing Analysis (Run 3)

The transition from the Run 2 configuration to the Run 3 configuration required a coordinated adjustment of two key parameters: the reduction of the penalization threshold λ and the concurrent decrease in the descent step size (ν for PGD-based algorithms and μ for PAM-based schemes). In the Run 2 parameterization, the values of λ were excessively high (up to 1.5) and disproportionate relative to step sizes that were still overly large or unbalanced. This imposed an excessive

constraint during the soft-thresholding step, leading to slow convergence and the estimator inability to fully resolve active attack support.

Reducing λ alone would have made the algorithms prone to numerical overshooting due to overly large descent steps. Therefore, a Baseline Balancing configuration was established: the threshold λ was assigned within a moderate range $([0.5, 0.8])$, and concurrently, the step sizes were reduced ($\nu = 0.2$ for O-PGD and AO-PGD; $\mu = 0.5$ for O-PAM and AO-PAM).

Table III summarizes the coordinated parameter recalibration.

TABLE III: Parameter Configuration for Baseline Balancing (Run 3)

Algorithm	μ/ν	λ
O-PGD	0.2	0.8
O-PAM	0.5	0.5
AO-PGD	0.2	0.8
AO-PAM	0.5	0.8

1) *Error Classification (False Negatives and False Positives)*: The combined reduction of the threshold and the step size allowed the system to successfully recalibrate attack classification. Observing the False Negative profile (Fig. 9), all algorithms achieve asymptotic convergence to zero within the very first iterations ($k < 10$). Concurrently, the False Positive count (Fig. 10) demonstrates that the reduced λ threshold remains sufficiently high to ensure complete noise rejection. The algorithms reach zero misclassifications immediately, confirming the effectiveness of the new parameter balancing.

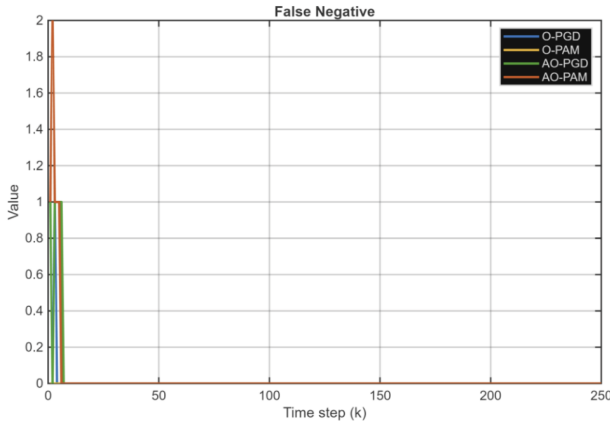


Fig. 9: Run 3 False Negative error across iterations.

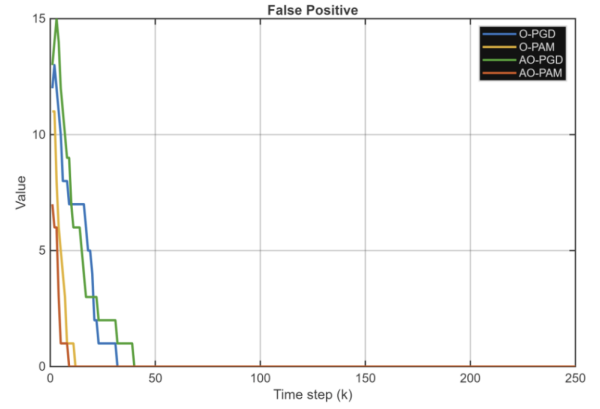


Fig. 10: Run 3 False Positive count across iterations.

2) *Support Attack Error*: The analysis of the Support Attack Error profile (Fig. 11) highlights the differential impact of the reduced parameters across the optimization schemes. The O-PAM and AO-PAM variants exhibit a rapid and monotonic decrease, driving the support discrepancy to zero in approximately 10 time steps. Conversely, the O-PGD and AO-PGD algorithms maintain a convergence behavior characterized by a distinct stepped trend, reaching zero near iteration $k = 40$. This confirms that a descent step $\nu = 0.2$ guarantees numerical stability, while requiring a larger iteration count to zero the uncorrupted components.

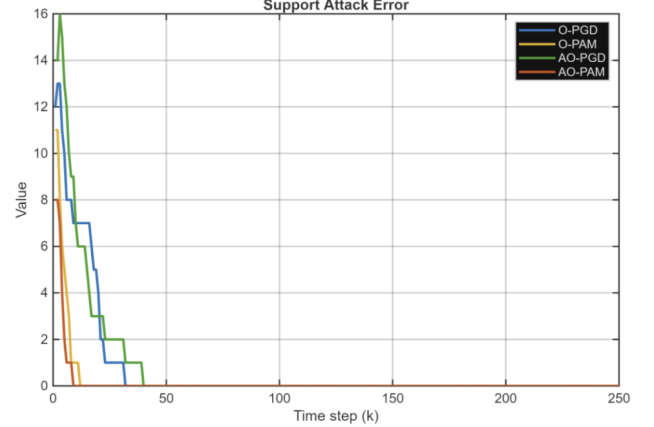


Fig. 11: Run 3 Support Attack Error across iterations.

3) *State Estimation Error*: The State Estimation Error profile (Fig. 12) confirms that the Run 3 parameters constitute a numerically stable configuration, but it demonstrates the structural limitations of operating with constant step sizes. Conservative ν and μ values eliminate divergence risks, but cause the estimation error to stabilize at constant asymptotic plateaus. The O-PAM algorithm achieves the quantitatively superior convergence rate, rapidly reaching a residual error on the order of 10^{-2} . The remaining three implementations exhibit slower convergence rates and higher residual tracking errors. This asymptotic saturation demonstrates that a fixed step size becomes ineffective in the immediate proximity of the local minimum.

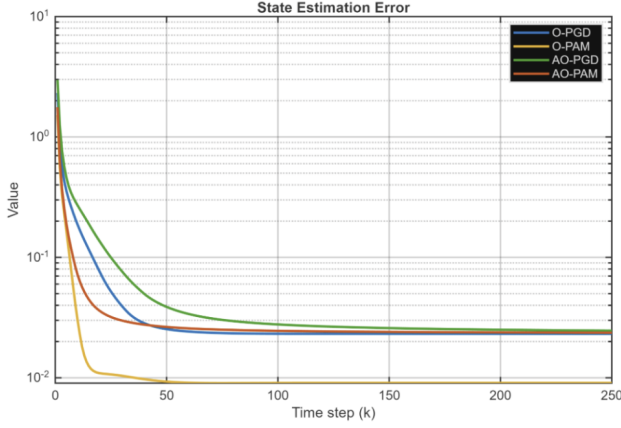


Fig. 12: Run 3 State Estimation Error across iterations.

4) *Summary of Baseline Balancing*: In summary, the parameter configuration of Run 3 resolves the classification failure modes observed in Run 2. The O-PGD and AO-PGD algorithms demonstrate high computational efficiency, achieving tracking performance quantitatively close to optimality with substantial numerical stability. Conversely, the AO-PAM architecture exhibits suboptimal tracking behavior, maintaining an asymptotic residual error higher than anticipated. This performance disparity confirms that constant step sizes are insufficient to maximize the estimation efficiency across all algorithmic variants, motivating the implementation of dynamic step-size scheduling in the final optimal configuration.

D. Optimal Configuration Analysis (Run 4)

The transition to the final experimental run defines the optimal system configuration, introducing targeted adjustments to the learning step sizes (ν and μ) and implementing a scheduled step-size decay mechanism based on the iteration threshold.

Specifically, the parameters for the PGD-based algorithms were assigned as $\nu = 0.2$ and $\lambda = 0.8$. For the PAM-based architectures, differentiated step-size values were adopted: $\mu = 1.0$ for O-PAM and an increased initial value $\mu = 1.2$ for AO-PAM, while maintaining $\lambda = 0.5$ for both. The primary structural modification consists of applying an iteration threshold at $k = 100$: every 100 iterations, the step size is halved for all algorithms, with the exception of O-PAM, whose fixed configuration is inherently stable. This parametric architecture aims to combine rapid initial transient convergence with high asymptotic tracking accuracy.

Table IV summarizes the optimal parameterization and decay scheduling.

TABLE IV: Parameter Configuration for Optimal Phase (Run 4)

Algorithm	Initial μ/ν	λ	Decay Schedule
O-PGD	0.2	0.8	Halved at $k = 100$
O-PAM	1.0	0.5	Constant (No decay)
AO-PGD	0.2	0.8	Halved at $k = 100$
AO-PAM	1.2	0.5	Halved at $k = 100$

1) *Error Classification (False Negatives and False Positives)*: The False Negative profile (Fig. 13) demonstrates high detection responsiveness: across all algorithmic variants, the error converges to zero within the initial iterations, guaranteeing that no physical attack injection is missed by the monitoring architecture.

Concurrently, the False Positive trajectory (Fig. 14) highlights the efficiency of the optimal parameterization. The O-PAM algorithm drives the uncorrupted channels to zero in approximately 15 time steps. O-PGD and AO-PAM reach zero misclassifications at iteration $k = 25$; specifically, assigning an increased initial step size to AO-PAM ($\mu = 1.2$) slightly dilated the transient phase, but ultimately ensured superior precision. Finally, AO-PGD requires a marginally longer processing window, converging to absolute zero near iteration $k = 45$.

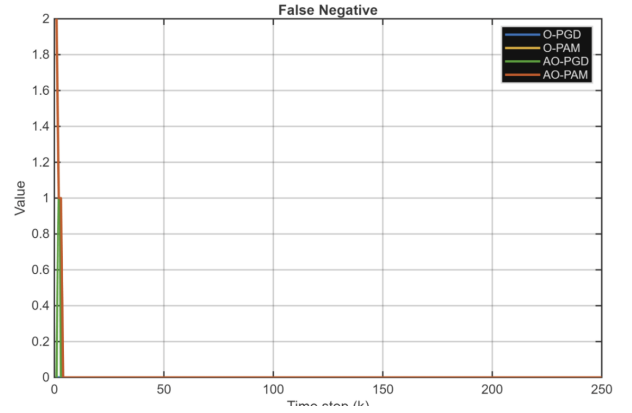


Fig. 13: Run 4 False Negative error across iterations.

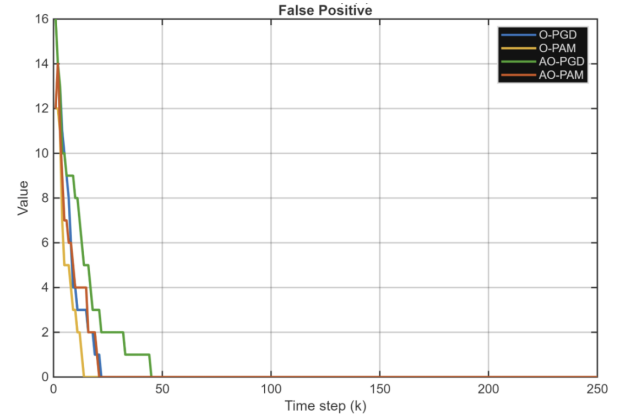


Fig. 14: Run 4 False Positive count across iterations.

2) *Support Attack Error*: The Support Attack Error profile (Fig. 15) directly mirrors the progressive elimination of False Positives. The O-PAM algorithm nullifies the support discrepancy in the shortest operational window, whereas AO-PGD completes exact support identification near time step $k = 45$. A fundamental analytical observation is that exact classification of the attack vector occurs for all evaluated methods strictly before the activation of the step decay mechanism ($k < 100$). This allows the subsequent optimization phase to operate over a fully stabilized support, concentrating computational resources exclusively on state error minimization.

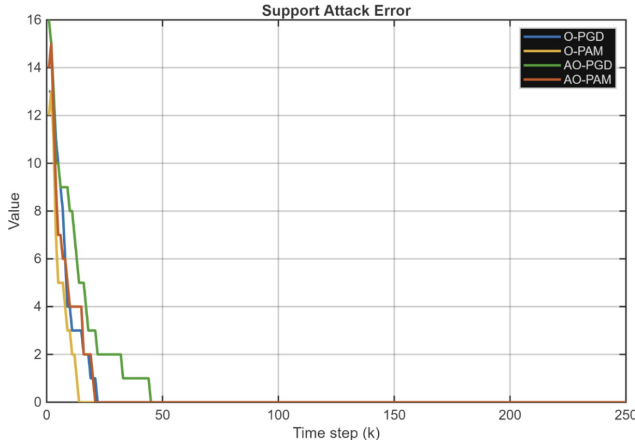


Fig. 15: Run 4 Support Attack Error across iterations.

3) *State Estimation Error*: The State Estimation Error profile (Fig. 16) unambiguously quantifies the numerical benefits derived from scheduled threshold decay. The O-PAM algorithm, operating with a constant step size, rapidly reaches its asymptotic tracking limit on the order of 10^{-2} . The O-PGD, AO-PGD, and AO-PAM architectures initiate convergence by exploiting their aggressive initial ν and μ values to achieve rapid transient error reduction. Upon crossing the threshold $k = 100$, step-size halving dampens the magnitude of state updates. This dynamic scheduling refines asymptotic convergence and suppresses numerical oscillations in the immediate vicinity of the optimum, aligning the residual tracking error of all algorithmic variants near 10^{-2} .

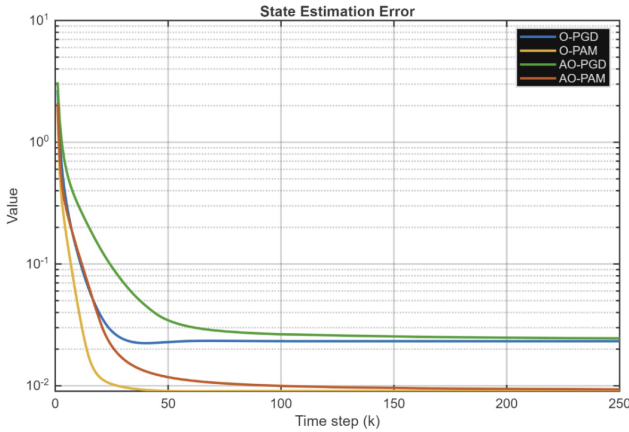


Fig. 16: Run 4 State Estimation Error across iterations.

4) *Summary of Optimal Configuration*: In conclusion, deploying an aggressive initial learning step—necessary to guarantee rapid support identification and overcome the transient underperformance of complex structures such as AO-PAM—coupled with scheduled step decay for final asymptotic stabilization, constitutes the definitive, optimal formulation for real-time Cyber-Physical System monitoring.

III. CONCLUSIONS

This paper evaluated the problem of Secure State Estimation (SSE) for marginally stable Cyber-Physical Systems (CPS) subject to sparse malicious sensor injections. Preliminary analytical assessments established that the presence of system

eigenvalues at unity ($\lambda = 1$) renders classical detection schemes, such as the standard Luenberger observer, mathematically ineffective. This structural limitation arises because the nominal unforced system dynamics overlap with the constant attack vector, causing the augmented system to yield an infinite set of feasible solutions.

To overcome this unobservability constraint, optimization-based online algorithms leveraging sparse recovery principles (O-PGD, O-PAM, AO-PGD, and AO-PAM) were implemented and evaluated. The experimental campaign demonstrated that parameter tuning governs both transient attack identification and asymptotic state tracking accuracy. Testing under uncalibrated conditions (Runs 1 and 2) highlighted two critical failure modes: excessive regularization thresholds induce severe over-shrinkage that blinds the monitoring architecture to active attacks, whereas inadequate descent step sizes generate slow, stepped error convergence profiles.

Systematic recalibration confirmed that while fixed-step configurations (Run 3) ensure numerical stability, they limit estimation performance by causing algorithms to stall at suboptimal asymptotic error plateaus. The optimal operating condition (Run 4) was ultimately achieved by introducing a dynamic step-size scheduling mechanism triggered at iteration $k = 100$. This architecture successfully decoupled the two primary estimation phases. First, aggressive initial step sizes guaranteed rapid support identification, driving False Negative and False Positive misclassifications to zero within the first 45 time steps. Second, halving the descent step upon crossing the iteration threshold suppressed numerical oscillations near the local optimum.

Ultimately, the results confirm that combining an aggressive initial search step with scheduled step-size reduction provides a robust framework for real-time CPS monitoring. Among the evaluated methods, O-PAM demonstrated superior quantitative efficiency, achieving the fastest support identification and converging to an asymptotic state tracking error on the order of 10^{-2} . Consequently, dynamic optimization observers successfully resolve the ambiguity of unobservable cyber-attacks, ensuring reliable online state estimation without requiring exhaustive offline data collection.